"Vision Transformers Need Registers": Token Norms, Register Ablation and Attention Maps on Imagenette

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Code: Vision-Transformers-need-registers

***Abstract*—Large Vision Transformers (ViTs) can develop high-norm outlier tokens in low-information image regions, degrading attention maps. Appending learnable "register" tokens mitigates this by giving the network a place to store global information. We implemented the register-token method end to end under a single-GPU budget. In a pre-trained DINOv2 ViT-g/14 we reproduced the artifact phenomenon: about 2.9% of final-layer patch tokens form a separate high-norm cluster, with norms up to about 11 times the median. We then built a compact ViT (2.76 M parameters) with a configurable number of registers and trained 0- and 4-register models on Imagenette over five seeds. Registers added only 768 parameters and gave a higher mean accuracy (77.11% versus 76.50% top-1; 76.61% versus 76.08% linear probe), ahead in four of five seeds, a consistent positive trend that remains within statistical noise at this sample size. Our results show that registers integrate cleanly into a ViT without hurting accuracy, and that the artifacts we measured appear in the giant pre-trained model but not in our small ViT, consistent with the scale dependence described in the paper.**

***Index Terms*—vision transformer, register tokens, attention maps, DINOv2, reproducibility**

I. Introduction

Vision Transformers (ViTs) [2] split an image into patches, embed each patch as a token, and process the token sequence, together with a `[CLS]` token, using self-attention. Self-supervised models such as DINO [4] and DINOv2 [3] produce features that transfer well, and their `[CLS]`-to-patch attention often highlights objects.

Darcet *et al.* [1] report that in sufficiently large models this picture is disturbed. A small fraction of patch tokens have a norm roughly an order of magnitude above the rest, and attention maps show isolated spikes on those tokens. According to the paper, these artifacts (i) appear only in sufficiently large models, (ii) emerge after sufficiently long training, (iii) arise mostly in middle-to-late layers, and (iv) sit on patches that are redundant with their neighbours, such as uniform background.

The proposed explanation is that the network has more capacity than it needs for local features, so it recycles low-information tokens to store global information, effectively using patch tokens as scratch memory. This hurts dense prediction and object discovery, and it yields noisy attention maps. It also exposes an architectural gap: the model has no dedicated place for intermediate global computation. *Registers* fill this gap. They are extra learnable tokens that are appended to the input, processed by every layer, and discarded at the output. A related phenomenon, attention sinks and massive activations, has been reported in language models [5], [6].

This paper reproduces the core experiments on a single Kaggle T4 GPU. Our contributions are:

- a norm analysis of the pre-trained DINOv2 ViT-g/14, in which we reproduce the high-norm artifact tokens described in the paper.
- a compact ViT with 0 or 4 register tokens, trained on Imagenette over five seeds, compared by accuracy, linear probe, and `[CLS]` attention maps.

II. Method

Let an image be split into N patches, each linearly embedded to $\mathbf{x}_i \in \mathbb{R}^D$. A standard ViT builds the input sequence

$$\mathbf{Z}_0 = [\mathbf{x}_{\text{cls}}; \mathbf{x}_1; \ldots; \mathbf{x}_N] + \mathbf{E}_{\text{pos}}, \quad \mathbf{Z}_0 \in \mathbb{R}^{(1+N)\times D}. \tag{1}$$

Adding registers. We introduce R learnable vectors $\mathbf{r}_1, \ldots, \mathbf{r}_R \in \mathbb{R}^D$ shared across all images. They are appended *after* the positional embedding is added, so they carry no spatial position:

$$\tilde{\mathbf{Z}}_0 = [\mathbf{Z}_0; \mathbf{r}_1; \ldots; \mathbf{r}_R] \in \mathbb{R}^{(1+N+R)\times D}. \tag{2}$$

The extended sequence passes through L pre-norm transformer blocks:

$$\mathbf{Z}'_\ell = \mathbf{Z}_{\ell-1} + \text{MSA}(\text{LN}(\mathbf{Z}_{\ell-1})), \tag{3}$$

$$\mathbf{Z}_\ell = \mathbf{Z}'_\ell + \text{MLP}(\text{LN}(\mathbf{Z}'_\ell)), \tag{4}$$

where self-attention is $\text{softmax}(\mathbf{Q}\mathbf{K}^\top/\sqrt{d_h})\mathbf{V}$ over all $1 + N + R$ tokens. Registers therefore read from and write to the image tokens in every layer, but no loss term supervises them directly.

Discarding registers. After the last block the registers are removed and only the original tokens are used:

$$\mathbf{Z}_L^{\text{out}} = \mathbf{Z}_L[0{:}1{+}N], \quad \hat{\mathbf{y}} = \mathbf{W}\,\text{LN}\big(\mathbf{Z}_L^{\text{out}}[0]\big) + \mathbf{b}. \tag{5}$$

The extra cost is R tokens per layer, about 1.6% more tokens for R=4 and N=256, and $R \cdot D = 768$ extra parameters for D=192.

Artifact measurement. For final-layer patch tokens $\mathbf{z}_i$ we use $\|\mathbf{z}_i\|_2$. A token is counted as an outlier if $\|\mathbf{z}_i\|_2 > 150$ (the

absolute threshold used in [1] for DINOv2 ViT-g) or $\|\mathbf{z}_i\|_2 > 2.5 \cdot \mathrm{median}_j \|\mathbf{z}_j\|_2$ within the same image (a relative threshold that does not depend on a model's norm scale).

III. Implementation Details

All experiments use PyTorch 2.11 (CUDA 12.8) on a Kaggle notebook with one NVIDIA Tesla T4 GPU. Table I lists how the setting was scaled down to fit this budget.

Data pipeline. Imagenette [7] images are decoded once, resized and center-cropped to 144 px, and kept as `uint8` tensors on the GPU (caching took a few seconds). Training uses a random 128 px crop and horizontal flip performed on the GPU; validation uses a center 128 px crop.

Training. We use AdamW with learning rate 10^{-3} and weight decay 0.05 (none on biases, norms, `[CLS]`, positional and register parameters), 3 warm-up epochs followed by cosine decay, label smoothing 0.1, batch size 128, gradient clipping at 1.0, and fp16 autocast for 40 epochs. Registers are initialised from a truncated normal (σ=0.02) and have no positional embedding. Fused scaled-dot-product attention was used for training; attention weights for analysis were obtained with an explicit float32 softmax in the last block.

Part 1 (pre-trained DINOv2). Token norms are taken from the output of the final block, before the last LayerNorm, of `dinov2_vitg14` loaded through `torch.hub` [3]. We use 64 randomly chosen validation images (16,384 patch tokens) resized to 224 px; since the dataset images are at most 160 px on the short side, they are upsampled.

Linear probe. Frozen `[CLS]` features (after the final LayerNorm) are standardised, and a single linear layer is trained for 300 full-batch AdamW steps on the training features and evaluated on the validation set.

IV. Results versus the Paper

A. Token norms in pre-trained DINOv2

Fig. 1 shows the distribution of final-layer patch-token norms of the pre-trained DINOv2 ViT-g/14 (no registers) over 64 validation images. The distribution is clearly bimodal: most tokens form a compact group with a median norm of 49.1, while a separate group of tokens reaches norms from roughly 190 up to 543. About 2.9% of the tokens (about 470 of 16,384) belong to this high-norm group. Both outlier definitions agree: 2.88% of tokens exceed the absolute threshold of 150, and the same fraction exceeds 2.5× the median of their own image. The largest norm is about 11 times the median, i.e. an order of magnitude above ordinary tokens, which reproduces the artifact phenomenon described in [1].

Fig. 2 shows where these tokens sit on the four validation images with the most outliers. They are isolated tokens, mostly in background regions such as grass, sky and plain backdrops, in line with the location pattern reported in the paper.

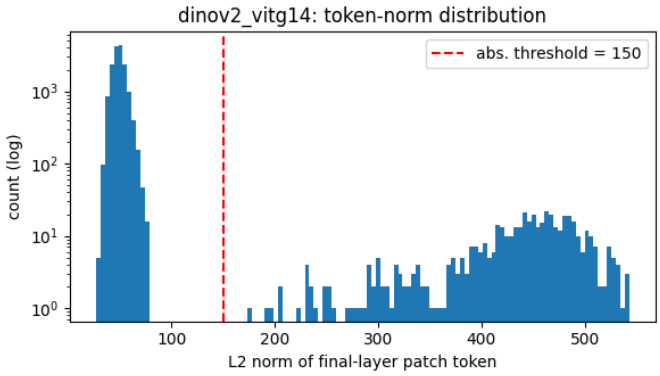


Fig. 1. Final-layer patch-token L2 norms of DINOv2 ViT-g/14 without registers over 64 validation images (log count). A separate high-norm group appears beyond the absolute threshold of 150 (dashed line).

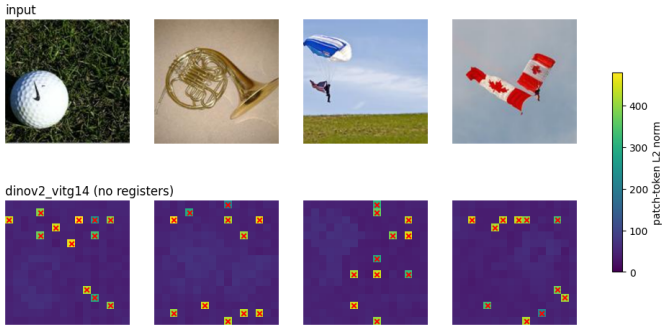


Fig. 2. Patch-token L2 norm maps (16×16) of DINOv2 ViT-g/14 without registers on the four validation images with the most outliers (top row: inputs). Red crosses mark tokens above 2.5× the image median.

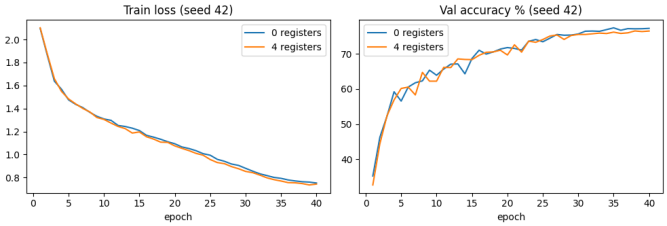


Fig. 3. Training loss and validation accuracy for seed 42, with 0 and 4 registers.

B. Accuracy of the small ViT

Table II reports validation accuracy as mean ± standard deviation over five seeds (per-run values in the repository). The training curves of the two models are close (Fig. 3).

The register model reached a higher mean accuracy on both measures. Per-seed differences (4 minus 0 registers) are −0.76, +0.26, +0.76, +2.09, +0.66 points for supervised accuracy and −1.14, +0.23, +1.22, +1.45, +0.91 for the linear probe, so registers were ahead in four of the five seeds in both cases. The mean gains of 0.60 and 0.54 points are of similar size to the seed standard deviations; an unpaired Welch test gives $p = 0.12$ and $p = 0.19$ (paired tests: $p = 0.26$ and $p = 0.32$). We therefore read the result as a consistent small positive trend that five seeds cannot yet separate from run-to-run noise. This agrees with the paper's claim that registers do not degrade

TABLE I
SCALE-DOWN RELATIVE TO THE ORIGINAL PAPER [1]

Aspect	**Paper**	**This work**
Data	ImageNet-1k and large pre-training sets	Imagenette (`imagenette2-160`), 10 classes, 9,469 train / 3,925 val images
Model	DINOv2 ViT-g/14 (about 1B parameters) and other large ViTs	Part 1: pre-trained DINOv2 ViT-g/14. Parts 2–3: custom ViT, patch 8, width 192, depth 6, 3 heads (2.76 M parameters)
Input	224 px	Part 1: 224 px. Small ViT: 128 px, 256 patch tokens
Training	Self-supervised / long schedules	Supervised from scratch, 40 epochs, AdamW, cosine schedule, fp16
Registers	4 (sweep over counts)	0 vs. 4
Evaluation	ImageNet probe, dense tasks, object discovery	Validation accuracy, linear probe, attention and norm maps
Seeds	Single large runs	42, 43, 44, 45, 46

TABLE II
IMAGENETTE VALIDATION ACCURACY (%), MEAN ± STD OVER 5 SEEDS

Model	Supervised top-1	Linear probe
Small ViT, 0 registers	76.50 ± 0.57	76.08 ± 0.41
Small ViT, 4 registers	77.11 ± 0.52	76.61 ± 0.71
Difference (4 − 0)	$+0.60$	$+0.54$

TABLE III
ARTIFACT AND ATTENTION STATISTICS (SEED 42, 1000 VAL IMAGES)

Model	Outliers	Max/med.	Entropy	To regs.
0 registers	0.0000	1.759	0.744	n/a
4 registers	0.0001	1.702	0.777	0.0065

classification accuracy [1]; we do not evaluate the dense tasks or object discovery on which the paper reports larger benefits.

C. Attention maps and artifact metrics

Fig. 4 compares last-layer `[CLS]` attention (mean over heads) and patch-token norms of the two small models on the same validation images, and Table III summarises 1000 validation images for the seed-42 models.

Columns: *Outliers* is the share of patch tokens above 2.5× the image median norm; *Max/med.* is the per-image ratio of the largest to the median patch norm, averaged over images; *Entropy* is the normalised entropy of the `[CLS]`-to-patch attention (1 means uniform); *To regs.* is the share of last-layer `[CLS]` attention that falls on registers.

Observations. Neither small model developed high-norm artifacts: the outlier fraction is essentially zero for both and the max/median norm ratio stays below 1.8, far from the roughly 11 times seen in ViT-g/14. With registers the max/median ratio is slightly lower (1.702 versus 1.759) and the `[CLS]` attention is somewhat more spread across patches (normalised entropy 0.777 versus 0.744). The registers receive 0.65% of the last-layer `[CLS]` attention, a small share that shows they are used lightly by this model. These statistics come from a single seed, so we treat them as indicative.

V. ABLATION AND FAILURE CASES

With versus without registers. The two models differ only in 768 extra parameters and four extra tokens per layer, so differences in behaviour are attributable to registers, subject to seed noise. Registers gave a higher mean accuracy in both measures (Table II) and a slightly lower norm spread with more evenly spread attention (Table III). Since the small baseline has no artifacts, this ablation shows that registers can be added at negligible cost without harm, while artifact removal itself is best examined in a model that exhibits artifacts, such as ViT-g/14.

Too many registers. We did not sweep the number of registers, so we have no measured evidence on this failure case. Plausible failure modes, stated as hypotheses, are: (i) sequence length and compute grow by $R/(1+N)$ per layer, which matters for short sequences (with 64 patch tokens, R=16 adds 25%); (ii) registers compete with `[CLS]` and patches for attention mass and could dilute `[CLS]` attention on small images; (iii) on a tiny dataset, extra tokens and parameters can increase over-fitting when there is nothing to repair. The paper reports that a small number of registers is enough [1]. A sweep over $R \in \{0, 1, 2, 4, 8, 16\}$ is a straightforward extension.

Training length and model size. According to [1], outliers appear only after a substantial part of training and only above a model-size threshold. Our measurements fit this picture: the giant ViT-g/14 shows a clear outlier group, while our 2.76 M-parameter ViT trained for 40 epochs on fewer than 10,000 images does not. We did not vary model size or training length within a controlled experiment, so this is a consistency check with the paper rather than a test of its claim. Training larger backbones for longer, ideally with self-supervision and dense evaluation, is the natural next step.

Other limitations.

1) The linear probe uses features of a supervised backbone, whereas the paper probes self-supervised features.
2) Imagenette has 10 classes, so accuracies saturate and differences are small.

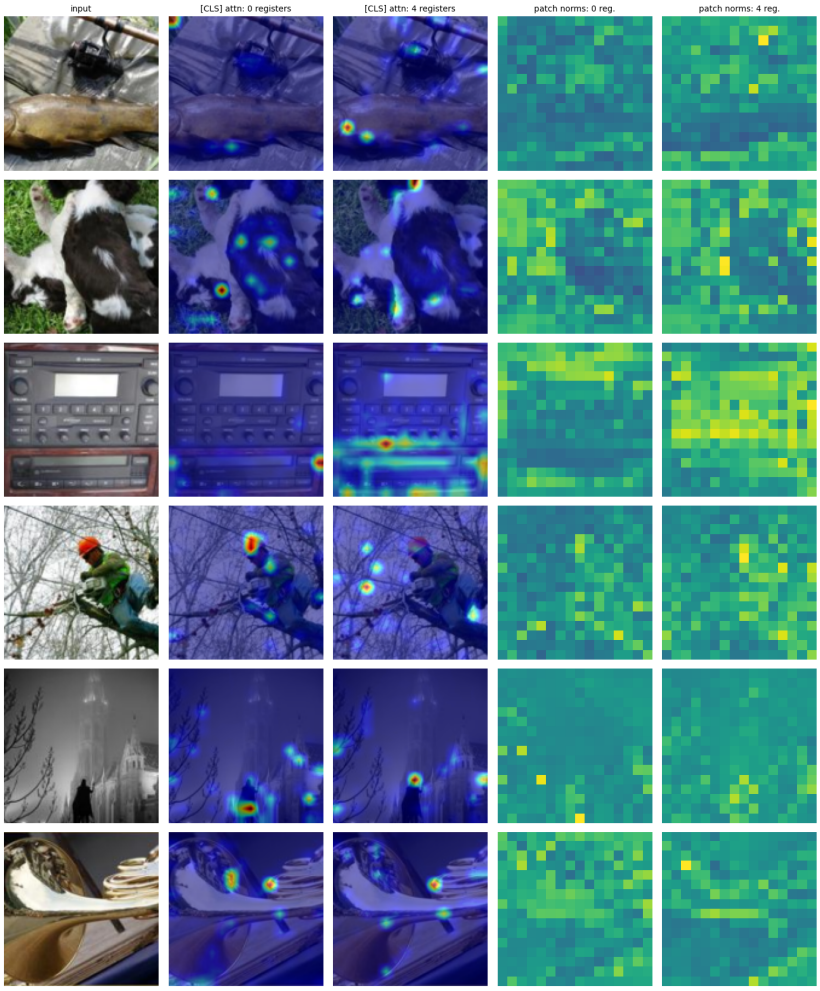


Fig. 4. Last-layer `[CLS]` attention and patch-token norms of the small ViT with 0 and 4 registers (seed 42) on six validation images.

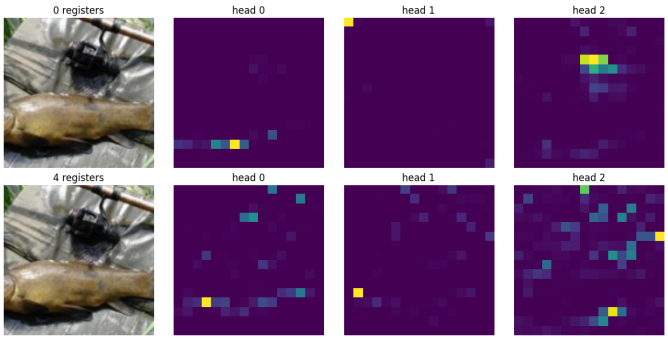


Fig. 5. Per-head last-layer `[CLS]` attention for one image, without (top) and with (bottom) registers.

3) Five seeds give a rough noise estimate; the accuracy difference is not statistically significant (Welch $p > 0.1$).
4) GPU kernels were not forced to be deterministic, so repeated runs can differ slightly.
5) Part 1 used 64 images and one measurement point.
6) Attention maps and Table III use the seed-42 models only.

VI. Conclusion

We implemented the register-token method end to end under a single-GPU budget. In the pre-trained DINOv2 ViT-g/14 we reproduced the artifact phenomenon: a separate group of high-norm patch tokens (2.9% of tokens, up to about 11 times the median norm), mostly on background regions, while our small ViT, as expected at its size, did not. We then built a compact ViT with registers and found that four registers cost only 768 parameters and gave a higher mean accuracy (+0.60 supervised and +0.54 linear-probe points over five seeds, ahead in four of five seeds), a consistent positive trend that is not yet statistically significant.

The results support the claim that registers are a cheap modification that does not hurt accuracy, and they match the paper's observation that artifacts emerge at large model scale. Testing the artifact-removal benefit directly requires larger register-equipped models, longer training and dense evaluation. Code and results are available at Vision-Transformers-need-registers.

References

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